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PROJECT NAME:

SCHOOL MANAGEMENT SYSTEM

PROJECT OVERVIEW :-

The **School Management System** is a comprehensive software solution designed to streamline the administration of student records and academic performance. The system provides a secure and organized environment for administrators to manage data while allowing students (users) easy access to their results.

Core Functionalities

The application is divided into two primary modules, ensuring a clear separation of concerns between administrative control and user access:

1. Administrative Module

The Admin panel acts as the command center of the system, offering ten specialized tools for total data control:

- **Data Entry & Maintenance:** Add new student records, delete specific entries, or reset the database using the "Delete All" feature.
- **Academic Management:** Update student marks and search for specific student profiles instantly.
- **Analytics & Reporting:** View the total number of admissions, calculate percentages, and sort the student roster based on academic performance (marks).
- **Navigation:** A seamless "Exit to Main Menu" option ensures smooth workflow transitions.

2. User Module

The User interface is designed for simplicity and efficiency, focusing on information retrieval:

- **Result Access:** Students can view their individual academic results.
 - **Merit Recognition:** A "Class Topper" feature allows users to identify the highest-achieving student in the batch.
 - **Ease of Use:** An intuitive exit option to return to the landing page.
-

: FUNCTIONAL REQUIREMENTS OF PROJECT :

1. Administrative Functional Requirements

As the primary controller of the system, the **Admin** module must satisfy the following requirements:

- **Student Registration:** The system shall allow the Admin to create and add new student profiles to the database.

- **Record Modification:** The system shall provide functionality to update academic marks for existing students.
- **Data Removal:** The system shall allow the Admin to delete a specific student's record using a unique identifier.
- **Database Reset:** The system shall provide a "Delete All" option to wipe the entire database for a new academic session.
- **Data Retrieval:**
 - **Search:** The system shall allow the Admin to search for a specific student's details.
 - **Bulk View:** The system shall display a complete list of all students currently enrolled.
- **Analytical Reporting:**
 - **Performance Calculation:** The system shall automatically calculate the percentage for each student.

- **Merit Sorting:** The system shall provide a feature to sort the entire student list based on marks achieved.
 - **Enrollment Statistics:** The system shall display the total count of active admissions in the system.
 - **Navigation:** The system shall allow the Admin to return to the Main Menu at any time.
-

2. User / Student

Functional Requirements

As a general viewer (Student/Parent), the **User** module must satisfy the following requirements:

- **Result Access:** The system shall allow the User to view academic results and marks.
- **Achievement Tracking:** The system shall identify and display the "Class Topper" based on the highest marks recorded in the database.

- **Restricted Access:** The system shall ensure the User cannot modify, add, or delete any data (Read-Only access).
- **Session Termination:** The system shall provide a clear exit path for the User to return to the main portal or close the application.

: WIREFRAMES OF THE PROJECT:

1.

```
D:\Semester 01\PF\PFL\week 10>c++ school.cpp -o school.exe
D:\Semester 01\PF\PFL\week 10>school.exe
-----
-----WELCOME TO SCHOOL SYSTEM-----
-----
1. Enter as ADMIN
2. Enter as User/Student (to view only)
3. To EXIT The System
Please enter your choice(1-3):|
```

This is the first interface of our project where we can see the three options as it given

- 1) For Admin
- 2) For User/Student
- 3) For Closing

2.

```
3. TO EXIT THE SYSTEM
Please enter your choice(1-3):1
-----ADMIN SECTION-----
VERIFICATION BY PASS
Enter User name :admin
Enter Password:123

---WelCome TO ADMIN PANAL---
1. Add student record
2. View all student
3. Delete student
4. Update marks
5. Search students
6. Sort by marks
7. Calculate class average
8. Total number of admissions
9. DELETE all
10.Return to Main Manu
Please enter your choice :|
```

If the user selects option 1 and enter as admin then he/she required to enter the correct user name and password to get access to the admin block where he/she performs multiple tasks as you can se in the given picture.

3.

```
10. Return to Main Menu  
Please enter your choice :1  
Enter name of the student :ali  
Enter roll number of the student :1  
Enter marks of the student :990  
---Data is save---
```

When the admin presses 1 after getting access then the program show the interface to add some student .

4.

```
2. View all student  
3. Delete student  
4. Update marks  
5. Search students  
6. Sort by marks  
7. Calculate class average  
8. Total number of admissions  
9. DELETE all  
10. Return to Main Menu  
Please enter your choice :2  
---student list---  
1.ali Roll no. 1 marks 990  
2.umar Roll no. 2 marks 1000
```

When he/she presses 2 then the program show the list of all the students.

5.

```
10. Return to Main Menu  
Please enter your choice :3  
Delete student from record  
Enter the roll no of the student you want to delete :2  
---Record Deleted---
```

When admin enter no 3 then he can delete the registered student by roll number.

6.

```
10. Return to Main Menu  
Please enter your choice :4  
Enter the roll no. you want to update marks :1  
Previous Marks :1010  
Enter Updated Marks :1000  
---Marks updated---
```

When the admin enter the option 4 then he can update the mark of the student by roll number.

7.

```
Please enter your choice :5  
Enter the roll no. of the student you want to search : 1  
Student name :ali' Marks obtained :1000
```

When the admin selects 5 then he/she can search student by roll number.

8.

```
---WelCome TO ADMIN PANAL---  
1. Add student record  
2. View all student  
3. Delete student  
4. Update marks  
5. Search students  
6. Sort by marks  
7. Calculate class average  
8. Total number of admissions  
9. DELETE all  
10.Return to Main Manu  
Please enter your choice :6  
Students are arranged in assending oredr according to roll no !
```

When admin selects option 6 then the students are arrange according to their marks obtained.

9.

```
10.Return to Main Manu  
Please enter your choice :8  
Total student registered :2
```

When admin selects 8 then he/she can see the total number of students.

10.

```
Please enter your choice :9  
All data is cleared
```

When the admin selects the option then the all data that is stored is deleted.

11.

```
10.Return to Main Menu  
Please enter your choice :10  
---Return to Main Menu---  
  
-----  
-----WELCOME TO SCHOOL SYSTEM-----  
-----  
  
1. Enter as ADMIN  
2. Enter as User/Student (to view only)  
3. To EXIT The System  
Please enter your choice(1-3):|
```

When the admin selects 10 then the admin block is closed and he/she came back to the main interface of the program.

12.

```
-----  
-----WELCOME TO SCHOOL SYSTEM-----  
-----  
1. Enter as ADMIN  
2. Enter as User/Student (to view only)  
3. To EXIT The System  
Please enter your choice(1-3):2  
  
-----USER MENU-----  
1. View Result By Roll Number  
2. View Highest Marks  
3. Exit to MAIN MENU  
Enter your choice :|
```

Now If the person selects the option 2 In the main interface then he entered in in the user menu here he can selects the more three options.

1. View the result by roll number
2. View highest marks
3. To Exit to the main manu

13.

```
Enter your choice :1  
----View Your Result----  
Please Enter your Roll number :2  
Your name is mahmoodyou obtained 1055Marks
```

When the user enter the 1 option then he can search the marks by the roll number.

14.

```
Enter your choice :2
---Highest Marks Obtained---
Highest Marks obtained in class is 1060
```

When the user selects the option 2 then he see the highest marks of the student in the class.

15.

```
Enter your choice :3
---Exiting From Student Menu---

-----
-----WELCOME TO SCHOOL SYSTEM-----
-----

1. Enter as ADMIN
2. Enter as User/Student (to view only)
3. To EXIT The System
Please enter your choice(1-3):|
```

When the user selects the option 3 then he came back to the main menu.

: DATA STRUCTURE OF THE PROJECT :

I use these arrays in the given project

```
#include <iostream>
using namespace std;
main()
{
    int maxstudent = 100;
    string names[maxstudent]={"ali","aminaa","ammar","hafsa"};
    int rollnumber[maxstudent]={1,3,4,5};
    float marks[maxstudent]={1009,1000,1001,1054};
    string username, password,name="admin",passcode="123";
    int mainchoice, adminchoice, userchoice, currentcount = 0;
    bool keeprunning = true;
    // program will run until user himself want to exit
    while (keeprunning)
    {
        system("cls");

        cout
            << endl
            << "-----" << endl;
        cout << "-----WELCOME TO SCHOOL MANAGEMENT SYSTEM-----" << endl;
        cout << "-----" << endl;
        cout << "1. Enter as ADMIN" << endl;
        cout << "2. Enter as User/Student (to view only)" << endl;
        cout << "3. To EXIT The System" << endl;
        cout << "Please enter your choice(1-3):";
        cin >> mainchoice;

        if (mainchoice == 1)
        {
            // admin section
            system("cls");

            cout << "-----ADMIN SECTION-----" << endl;
```

: ALL THE CODE OF THE PROJECT IS :

```
#include <iostream>

using namespace std;

main()
{
    int maxstudent = 100;

    string
names[maxstudent]={"ali","aminaa","ammar","haf
sa"};

    int rollnumber[maxstudent]={1,3,4,5};

    float
marks[maxstudent]={1009,1000,1001,1054};

    string username,
password,name="admin",passcode="123";

    int mainchoice, adminchoice, userchoice,
currentcount = 0;
```

```
bool keeprunning = true;

// program will run until user himself want to
exit

while (keeprunning)
{
    system("cls");

    cout
        << endl
        << "-----"

" << endl;

    cout << "-----WELCOME TO SCHOOL
MANAGEMENT SYSTEM-----" << endl;

    cout << "-----"
----" << endl;

    cout << "1. Enter as ADMIN" << endl;
```



```
    cout << "2. Enter as User/Student (to view  
only)" << endl;
```

```
    cout << "3. To EXIT The System" << endl;
```

```
    cout << "Please enter your choice(1-3):";
```

```
    cin >> mainchoice;
```

```
    if (mainchoice == 1)
```

```
    {
```

```
        // admin section
```

```
        system("cls");
```

```
        cout << "-----ADMIN SECTION-----" <<  
endl;
```

```
        cout << "VERIFICATION BY PASS" << endl;
```

```
        cout << "Enter User name :";
```

```
cin >> username;

cout << "Enter Password:";

cin >> password;

if (name == username && passcode ==
password)
{
    // after enterining in admin
    bool admin = true;
    while (admin)
    {

        // admin code
        system("cls");

        cout << endl
            << endl
```

```
<< "---WelCome TO ADMIN PANAL---"
<< endl;

cout << "1. Add student record" << endl;
cout << "2. View all student " << endl;
cout << "3. Delete student " << endl;
cout << "4. Update marks" << endl;
cout << "5. Search students " << endl;
cout << "6. Sort by marks" << endl;
cout << "7. Calculate class average" <<
endl;

cout << "8. Total number of admissions"
<< endl;

cout << "9. DELETE all " << endl;
cout << "10.Return to Main Manu" <<
endl;

cout << "11. Update your User name
and Password";
```

```
cout << "Please enter your choice :";  
cin >> adminchoice;  
  
// choices starts from here  
if (adminchoice == 1)  
{  
    // Add student  
  
    system("cls");  
  
    if (currentcount < maxstudent)  
    {  
  
        cout << "Enter name of the student  
:";  
  
        cin >> names[currentcount];  
        cout << "Enter roll number of the  
student :";
```

```
        cin >> rollnumber[currentcount];
        cout << "Enter marks of the student
        :";

        cin >> marks[currentcount];
        currentcount++;
        cout << "---Data is saved---" <<
endl;
    }
}
else if (adminchoice == 2)
{

    system("cls");

    // view all students

    cout << "---student list---" << endl;
```

```
        cout<<"Name\tRollnumber\tMarks"<<
endl;

        for (int i = 0; i < currentcount; i++)
        {
            cout <<names[i] << "\t " <<
rollnumber[i] << "\t\t" << marks[i] << endl;
        }
    }
```

```
else if (adminchoice == 3)
{
```

```
    // Delete any student from record
    system("cls");
```

```
    cout << "Delete student from record"
<< endl;
```

```
cout << "Enter the roll no of the  
student you want to delete :";
```

```
int roll;
```

```
cin >> roll;
```

```
bool found = false;
```

```
for (int i = 0; i < currentcount; i++)
```

```
{
```

```
    if (rollnumber[i] == roll)
```

```
    {
```

```
        for (int j = i; j < currentcount - 1;
```

```
j++)
```

```
        {
```

```
            names[j] = names[j + 1];
```

```
            rollnumber[j] = rollnumber[j + 1];
```

```
            marks[j] = marks[j + 1];
```

```
        }
```

```
        currentcount--;  
        found = true;  
        cout << "---Record Deleted---" <<  
endl;  
        break;  
    }  
}  
if (found == false)  
{  
    cout << "Roll Number Not Found !"  
<< endl;  
}  
}
```

```
// exit from admin
```

```
else if (adminchoice == 10)
```



```
{

    // exiting from admin
    cout << "---Exiting to the main menu--
-";

    admin = false;
    break;
}

//update password

else if(adminchoice==11){

    system("cls");

    string newUsername,newPassword;
```

```
cout<<"Enter your previous user :";  
cin>>username;  
cout<<"Enter your previous password  
:";  
  
cin>>password;  
  
if(username=="admin" &&  
password=="123"){  
    cout<<"enter new user name : ";  
    cin>>newUsername;  
    cout<<"enter new password : ";  
    cin>>newPassword;  
    name=newUsername;  
    passcode=newPassword;  
  
}  
  
}
```

```
else if (adminchoice == 4)
```

```
// update marks of the student
```

```
{
```

```
    system("cls");
```

```
        cout << "Enter the roll no. you want to  
update marks :";
```

```
        int roll;
```

```
        int m;
```

```
        bool found = false;
```

```
        cin >> roll;
```

```
        for (int i = 0; i < currentcount; i++)
```

```
{
```

```
            if (rollnumber[i] == roll)
```

```
        {
            cout << "Previous Marks :" <<
marks[i] << endl;

            cout << "Name of student :" <<
names[i] << endl;

            cout << "Enter Updated Marks :";
            cin >> m;
            marks[i] = m;
            found = true;
            cout << "---Marks updated---";
            break;
        }
    }
    if (found == false)
    {
        cout << "Roll no not found !!" <<
endl;
```

```
}
```

```
}
```

```
else if (adminchoice == 5)
```

```
// search by roll number
```

```
{
```

```
    system("cls");
```

```
        cout << "Ente the roll no. of the  
student you want to search : ";
```

```
        int roll;
```

```
        cin >> roll;
```

```
        bool found = false;
```

```
        for (int i = 0; i < currentcount; i++)
```

```
        {
```

```
            if (rollnumber[i] == roll)
```

```
    {  
        cout << "Student name :" <<  
names[i] << " Marks obtained :" << marks[i];  
    }
```

```
found = true;
```

```
cout << "roll number not found ";
```

```
break;
```

```
}
```

```
if (found == false)
```

```
{
```

```
    cout << "Student not found !!";
```

```
}
```

```
}
```

```
else if (adminchoice == 6)
```

```
// sort by marks
```

```
{
```

```
    system("cls");
```

```
    // Bubble sorting by marks
```

```
    for (int i = 0; i < currentcount - 1; i++)
```

```
    {
```

```
        for (int j = 0; j < currentcount - i - 1;
```

```
            j++)
```

```
        {
```

```
            if (marks[j] < marks[j + 1])
```

```
            {
```

```
                int temp = marks[j];
```

```
                marks[j] = marks[j + 1];
```

```
                marks[j + 1] = temp;
```

```
            }
```

```

        }
    }

    // Total students

    cout << "Students are arranged in
assending oredr according to roll no ! ";

    cout<<endl<<"Name\tRollnumber\tM
arks"<<endl;

    for (int i = 0; i < currentcount; i++)
    {
        cout <<names[i] << "\t  " <<
rollnumber[i] << "\t\t" << marks[i] << endl;
    }
}

else if (adminchoice == 7)

// class average

{

```



```
system("cls");

float total = 0;
float average;
for (int i = 0; i < currentcount; i++)
{
    total = total + marks[i];
}
if (currentcount > 0)
{
    average = total / currentcount;
    cout << "Average of the class marks
is " << average << endl;
}
}

else if (adminchoice == 8)

// total number of admissions
```

```
{  
    system("cls");  
  
    cout << "Total student registered :" <<  
currentcount << endl;  
}  
else if (adminchoice == 9)  
{  
    // delete all data  
  
    system("cls");  
    string n;  
    cout << "Enter --YES-- to confirm :";  
    cin >> n;  
  
    if (n == "YES")
```

```
{

    currentcount = 0;

    cout <<

"===== " << endl;

    cout << "=== All data is
cleared===" << endl;

    cout <<

"===== " << endl;

}

else

{

    cout << "---Not Confirm---";

}

}

}
```

```
    }  
    else  
    {  
        cout << "Incorrect user name or password  
!" << endl;  
        cout<<"press ENTER key to continue....";  
        cin.ignore();  
        cin.get();  
    }  
  
}  
  
// exit the system  
  
else if (mainchoice == 3)  
{
```

```
        cout << "---Exiting from the main system---"
<< endl;

        keeprunning = false;
    }
    else if (mainchoice == 2)
    {
        // user menu

        system("cls");

        bool student = true;
        while (student)
        {
            // user menu

            cout << endl

            << endl
```

```
<< "-----USER MENU-----" << endl;
cout << "1. View Result By Roll Number "
<< endl;

cout << "2. View Highest Marks" << endl;
cout << "3. View All student names " <<
endl;

cout << "4. Search student by names " <<
endl;

cout << "5. Exit to MAIN MENU" << endl;
cout << "Enter your choice :";
cin >> userchoice;

// user choice if 1 view result

if (userchoice == 1)
{
```

```

// view result by roll number
system("cls");

cout << "----View Your Result----" <<
endl;

int roll;

cout << "Please Enter your Roll number
:";

cin >> roll;

bool found = false;
for (int i = 0; i < currentcount; i++)
{
    if (roll == rollnumber[i])
    {
        cout << "Your name is " << names[i]
<< "you obtained " << marks[i] << "Marks" << endl;
    }
}

```

```
        found = true;
        break;
    }
}
if (found == false)
{
    cout << "----Result Not Found----" <<
endl;
}
}
```

```
// choice 2 highest marks
```

```
else if (userchoice == 2)
```

```
// highest marks
```

```
{
```



```
        system("cls");
        cout << "---Highest Marks Obtained---"
<< endl;

        int max;

        max = marks[0];

        for (int i = 1; i < currentcount; i++)
        {
            if (marks[i] > max)
            {
                max = marks[i];
            }
        }

        cout << "Highest Marks obtained in
class is " << max;

    }

    else if (userchoice == 5)
    {
```

```

        // exit to main menu

        cout << "---Exiting From Student Menu--
-" << endl;

        student = false;

    }

    else if (userchoice == 3)

        // names of all the students

    {

        system("cls");

        cout << "==== All students names
====";

        for (int i = 0; i < currentcount; i++)

        {

            cout << i + 1 << " Name is : " <<
names[i] << endl;

```

```
    }  
}  
else if (userchoice == 4)  
{  
  
    system("cls");  
  
    // seearch by name  
    cout << "Enter the name of the student  
you want to search :";  
    string n;  
    cin >> n;  
    bool found = false;  
    for (int i = 0; i < currentcount; i++)  
    {  
        if (names[i] == n)  
        {
```

```
        cout << "Name of the student : " <<
names[i] << "Marks obtain : " << marks[i] << " Roll
number is : " << rollnumber[i];
```

```
        found = true;
```

```
        break;
```

```
    }
```

```
}
```

```
if (found == false)
```

```
{
```

```
    cout << "--- Name not found ---";
```

```
}
```

```
}
```

```
else
```

```
        {  
            cout << "Your Choice is invalid please  
enter again !" << endl;  
        }  
    }  
}  
else  
{  
    cout << "Your choice is invalid please enter  
again !" << endl;  
}  
}  
}
```

